HIREFLOW AI — SYSTEM TESTING & QUALITY BLUEPRINT

The Definitive System Architecture, Verification Manual, Scoring Mathematics & QA Protocols

VERSION 4.0.0 | 106 SUITES PASSED (738 TESTS) | ZERO-EGRESS PRIVACY ENFORCED

1. Executive Summary & Verification Scorecard

HireFlow v4 is an offline-capable, client-side career decision and resume intelligence platform engineered specifically for engineering students and fresher graduates entering the Indian and global software hiring markets. Every engine operates deterministically inside the browser with zero external network egress.

Quality Dimension	Baseline (v3)	Current (v4 Delivery)	Verification Status
Automated Test Suites	75 test suites	**106 test suites**	**PASS (100% green)**
Total Passing Unit Tests	617 tests	**738 tests**	**PASS (0 failed)**
Initial Entry Chunk	15.68 MB (gzip: 1.85 MB)	**1.83 MB (gzip: 286 kB)**	**-88.3% Payload Drop**
Privacy / Network Safety	Standard	**0 external egress calls**	**Verified via AST Check**
DOM Injection Security	Legacy static CSP	**0 dangerous DOM sinks**	**Strict Build CSP Injected**
Mobile Responsiveness	Desktop primary	**360x800 & 390x844 audit**	**>=44px touch targets**

2. The 10 Prime Architectural Directives (Invariants)

P-1: Zero Fabrication / Zero Invented Claims

No fake placement statistics, made-up salaries, or invented recruiter claims. Unknown attributes are explicitly labeled `unverified` with `null` values.

P-2: Deterministic Pure Logic Engines

Core engines in `src/engine/` are 100% pure functions. No `Math.random()`, no `Date.now()`, no network access. Identical inputs produce bit-identical results.

P-3: Zero External Network Egress & Local-Only Privacy

All resume and JD parsing occurs locally in the browser. Zero analytics SDKs, zero remote logging, zero tracking pixels. Enforced by automated AST scanners.

P-4: Calibrated Scoring Pillars

ATS score calibrated to Indian fresher hiring patterns: Hard Skills 40%, Evidence Quality 25%, Fresher Readiness 20%, Format Safety 15%. Minimum baseline 25.0.

P-5: Honest Provenance & Auditing

Every national exam, certification syllabus, and salary tier carries verified provenance metadata, publication source, and freshness timestamps.

P-6: Strict Candidate Scope Discipline

All recruiter, interviewer, and enterprise BHR modules were permanently purged to ensure laser-focused depth solely for candidate empowerment.

P-7: Honest Naming & Rejection Reality

Zero claims of predicting company selection outcomes or "callback odds". Metrics are strictly termed "Match Alignment", "Evidence Depth", and "Coverage".

P-8: WCAG 2.1 AA Accessibility

Visible focus rings on interactive elements (`focus-visible:ring-2`), redundant icons with color coding, accessible dialog focus traps, and `aria-live` polite alerts.

P-9: Fault-Tolerant Storage & Corruption Quarantine

Resilient storage adapter catches quota limits cleanly, quarantines malformed JSON under `hireflow:v4:<key>_corrupt` , and prevents app crashes.

P-10: Secure Context & LAN Degraded Mode

Full graceful degradation when running on local Wi-Fi over plain HTTP: SHA-256 falls back to FNV-1a, UUID falls back to high-entropy counters.

3. Tool-by-Tool QA Specification & Test Manual

4.1 ATS Resume Roaster (/tools/resume-roaster)

- **Pillars Evaluated**: Hard Skills (40%), Evidence (25%), Fresher Readiness (20%), Format Safety (15%).
- **Character-Span Annotator (D16)**: Evaluates character spans across resume text:
  - **RED**: Weak action verbs ("assisted", "helped", "worked on").
  - **AMBER**: Run-on sentences (>35 words), missing metrics.
  - **GREEN**: Evidenced technical skills, quantifiable metric patterns.
- **Rewrite Applicator**: Click any highlight to view rewrite suggestions and apply updates with one click.

4.2 Dual-Role Job Description Comparison (/tools/jd-compare)

- **Side-by-Side Analysis**: Compares two JDs or evaluates one resume against two distinct roles.
- **Overlap Breakdown**: Displays shared technical requirements vs unique requirements.
- **Role Presets (A5)**: 7 Indian fresher role archetypes with branch-bridge guidance for non-CS engineers.

4.3 7-Stage Zero-Hallucination Resume Tailoring (/tools/tailor)

- **Truth Gate Enforcer**: Guarantees tailoring never invents unevidenced skills or accomplishments.
- **Formatting Quality Sentinel (D15)**: Scans for broken glyphs, widow/orphan lines (< 3 words), and 1-page length overflows before export.

4.4 Project Build Briefs & Set-Cover Bundler (/tools/build-briefs)

- **Greedy Set-Cover Bundler**: Computes the minimum set of projects covering all open skill gaps.
- **Scaffold Generator (D17)**: Generates starter repository ZIP in-browser (README, Mermaid diagram, tests).
- **Resume Injection**: Injects completed project bullet with `[METRIC]` placeholders directly into draft.

4.5 Credential & Opportunity Engine (/tools/credential-engine)

- **National Exam Eligibility**: Rule-based eligibility checker for GATE, ISRO ICRB, AWS SAA, CDAC.
- **iCalendar Generator (D18)**: Generates RFC 5545 `.ics` calendar file with exam dates and study milestones.

4.6 Career Trajectory Hub & What-If Simulator (/tools/career-path)

- **India Tech Ladders**: Leveling and compensation tracks for IT Services, Unicorns, GCCs, and Big Tech.
- **What-If Simulator (D20)**: Live sliders for CGPA, Tier, and Branch; reveals campus corporate cutoffs.

4.7 TalentLens Intelligence & Saved Roles (/tools/talentlens)

- **Saved Roles Store (D19)**: Bookmark up to 50 roles with local persistence.
- **Prefill to Matcher**: Sends selected role requirements directly into JD Compare with one click.

4.8 Grounded Outreach Generator (/tools/reach-out)

- **Deterministic Follow-Ups**: Generates Day 3, Day 7, and post-interview messages with `[FILL_IN]` tags.
- **Subject Line Analyzer**: Flags spam trigger words, subject length, and personalization density.

4. Mathematical & Algorithmic Scoring Inventory

Metric ID	Display Title	Unit	Mathematical Formula	Config Source
`overall-ats-score`	Resume ATS Score	/100	`(HardSkills*0.40) + (Evidence*0.25) + (Fresher*0.20) + (Format*0.15)`	`ATS_PILLAR_WEIGHTS`
`jd-match-score`	JD Match Score	%	`(∑ Weight * MatchCredit * EvidenceMult) / (∑ Weight) * 100`	`MATCH_MULTIPLIERS`
`hard-skills-coverage`	Technical Coverage	%	`(Evidenced Mandatory Skills / Total Mandatory Skills) * 100`	`REQUIREMENT_WEIGHTS`
`evidence-depth-score`	Evidence Depth	/100	`(∑ BulletTierMultipliers) / TotalBullets * 100`	`EVIDENCE_LEVELS`
`verb-impact-score`	Verb Impact Density	/100	`((Strong*1.15) + (Standard*1.0) - (Weak*0.3)) / TotalBullets * 100`	`VERB_MODIFIERS`
`demanded-skill-score`	Market Demand Score	pts	`(Mandatory*3.0 + Preferred*1.5 + Mentioned*1.0) / TotalSavedJDs`	`SavedJdLibrary`
`branch-bridge-match`	Branch Alignment	%	`(TransferableSkills.len + CoreSkills.len) / TotalCoreSkills * 100`	`role-presets.json`

5. Storage Specifications & Application Limits

Store Name	Storage Key Prefix	Enforced Cap Limit	Eviction / Guard Policy
Active Profile	`hireflow:v4:profile`	5 resume versions	Prompt to replace or delete when 6th added
Scan History	`hireflow:v4:scan_history`	60 scans	Oldest scan automatically evicted (Ring Buffer)
Saved JDs	`hireflow:v4:saved_jds`	40 JDs	Deduplicated via hashText(); cap prompt
Saved Roles	`hireflow:v4:saved_roles`	50 roles	Cap prompt when limit reached
User Feedback	`hireflow:v4:feedback`	500 notes	Oldest feedback evicted; zero scoring mutation
Brief Progress	`hireflow:v4:brief_progress`	30 briefs	Status tracking & resume ledger state

6. Quick QA Execution Cheatsheet

```
# Run entire automated test suite (106 suites, 738 tests)
npx vitest run

# Run security, AST egress & hostile payload tests
npx vitest run tests/security/

# Run WCAG accessibility tests
npx vitest run tests/a11y/ tests/unit/stage10/crossCuttingA11y.test.ts

# Verify zero unauthorized network egress
npm run privacy:check

# Full production build & bundle audit
npm run build
npx tsx scripts/check-bundle-size.ts

# Test production build in preview server
npm run preview
# Open http://localhost:4173 in Edge or Chrome
```